

Bridge Phase-2 (No-Local-Runtime) — Execution Plan, Verified Against Current Tree

Revision: 1 **Last modified:** 2026-06-15T00:00:00Z **Status:** DESIGN / READY-FOR-EXECUTION (read-only survey produced this; no production source edited). FACT-cited per §11.4.6. **Authority:** Operator mandate 2026-06-15 — “Do not use any local running models using llama.cpp, only LLMsVerifier obtained strongest models and created bridge ... for direct access to model capabilities!” Locked decisions: **D1** remove local runtimes entirely · **D2** forbid ALL local runtimes (llama.cpp + Ollama) · **R1** Obsolete the SSH/distributed worker path (§11.4.90) · **R2** require API keys everywhere (no offline/local fallback) · **D4/D5** top-1 + in-process fallback chain. Bridge facade = pkg/bridge (BestTranslator/Invoke/BestModel/ListVerified). **Supersedes the estimates in:** docs/design/LOCAL_RUNTIME_REMOVAL.md (Rev 1) — this doc carries the verified-against-current-tree counts/paths. The companion docs/design/LLMSVERIFIER_BRIDGE.md covers the bridge build itself. **Verified against tree at:** HEAD a5860b2 (2026-06-15). Every file:line below was confirmed in this read-only pass.

0. TL;DR for the conductor

- **Bridge EXISTS and builds:** pkg/bridge/bridge.go (+ bridge_test.go, export_test.go, util.go, data/). BestTranslator/Invoke/BestModel/ListVerified are implemented. cmd/model-bridge already imports it. **The Phase-0 prerequisite (bridge present) is MET.**
- **Drop-in verdict: GO, with ONE tracked prerequisite gap (P-1, below).** bridge.BestTranslator returns translator.Translator backed by the SAME *llm.LLMTranslator that llm.NewLLMTranslator returns today — so the 14 high-level NewLLMTranslator call sites redirect cleanly. The gap: two consumers depend on the narrower llm.LLMClient interface (which has GetProviderName()); translator.Translator does NOT expose GetProviderName(). Those two need an adapter (P-1).
- **Verified site counts (vs the draft's estimates):** | Class | Draft estimate | **Verified (this pass)** |
| NewLLMTranslator non-test call sites | 9 components | **14 call sites** across 12 files |
| llama.cpp non-test source files | ≈25 sites | **16 files** (≈25 lines) |
| Ollama non-test source files | ≈13 sites | **~12 files** (token spread; O1–O13 hold, see §1.2) |
| local-runtime worker configs | 5 | **8 configs** |
| local-runtime helper scripts | 4 | **15 scripts** in internal/scripts/ |
| SSH binaries with API path | “llama-only” | **CONFIRMED llama-only** (grep for NewLLMTranslator/VerifiedFactory/bridge. returned EMPTY) |

- **8-angle verdict: GO with conditions** (R1 operator-confirm distributed/SSH removal = already granted via R1; R2 no-key → honest SKIP, no silent fallback — the bridge already hard-errors with no keys, §1.4 below).

1. Verified current inventory (FACT, file:line)

1.1 The 14 NewLLMTranslator redirect call sites (THE primary redirect surface)

llm.NewLLMTranslator(config) (*LLMTranslator, error) is defined pkg/translator/llm/llm.go:217. Non-test callers (verified this pass):

#	Site	Component	Current default
1	cmd/cli/main.go:350	cmd/cli	provider flag (incl. ollama/llamacpp)
2	cmd/markdown-translator/main.go:314	markdown-translator	deepseek (API)
3	cmd/preparation-translator/main.go:175	preparation-translator	resolves *_API_KEY (API-only already)
4	cmd/unified-translator/main.go:469	unified-translator	API path
5	cmd/unified-translator/main.go:513	unified-translator	second provisioning site
6	pkg/api/batch_handlers.go:123	api batch	provider switch
7	pkg/api/batch_handlers.go:235	api batch	provider switch
8	pkg/api/handler.go:778	api createTranslator	provider list
9	pkg/coordination/multi_llm.go:171	coordination	per-provider
10	pkg/distributed/coordinator.go:297	distributed	local translator (R1)
11	pkg/distributed/coordinator.go:302	distributed	local translator fallback (R1)
12	pkg/grpc/core_translator.go:320	grpc	provider routing
13	pkg/grpc/core_translator.go:361	grpc	provider routing (2nd)
14	pkg/preparation/coordinator.go:98	preparation pkg	per-provider

#	Site	Component	Current default
15	pkg/verification/ polisher.go:124	verification	per-provider

Draft missed #3, #5, #11, #13, #14, #15. (15 rows; #4/#5 same binary, #6/#7 same file, #10/#11 same file, #12/#13 same file → 12 files, 15 call sites.)
 pkg/challenge_runner/provider_challenges.go:51 references the
 STRING TestNewLLMTranslator in a go test -run — NOT a call site;
 leave or update test name only.

1.2 llama.cpp — 16 non-test files (FACT)

cmd/cli/main.go, cmd/ebook-translator/ebook_translator.go, cmd/
 markdown-translator/main.go, cmd/ssh-translation/main.go, cmd/
 translate-ssh/main.go, cmd/unified-translator/main.go, pkg/api/
 batch_handlers.go, pkg/api/handler.go, pkg/challenge_runner/
 provider_challenges.go, pkg/distributed/coordinator.go, pkg/
 distributed/pairing.go, pkg/grpc/core_translator.go, pkg/grpc/proto/
 translator.pb.go (regen, do NOT hand-edit), pkg/translator/llm/
 llamacpp_provider.go, pkg/translator/llm/llamacpp.go, pkg/
 translator/llm/llm.go.

unified-translator local flag/branch lines (FACT): case "ssh" :304, case
 "llamacpp" :306, executeSSHTranslation :314,
 executeLlamaCppTranslation :447, hardcoded gguf :391, -ssh-host/-ssh-
 user/-ssh-password/-ssh-port :627-630, -llama-binary/-llama-
 model :634-635, case "ssh" key-check :692, case "llamacpp" key-check
 :697-699, help text :764-772.

1.3 Ollama — O1–O13 (FACT, verified)

ID	Site (file:line)	Action
O1	pkg/translator/llm/ ollama.go (whole file, OllamaClient, :11434 /api/ generate)	DELETE (§11.4.124 git-history first)
O2	pkg/translator/llm/ llm.go:34 ProviderOllama const	REMOVE
O3	pkg/translator/llm/ llm.go:70 ValidModels[ProviderOllama]	REMOVE
O4	pkg/translator/llm/llm.go “Ollama/LlamaCpp custom-model warn” arm	REMOVE (both gone)
O5	pkg/translator/llm/llm.go case ProviderOllama: factory branch	REMOVE; unknown → error

ID	Site (file:line)	Action
	(NewLLMTranslatorWithConfig, ~:251–312)	
O6	cmd/unified-translator/main.go:702 case "mock", "ollama": no-key arm	REMOVE ollama; case "ollama" → error/redirect
O7	cmd/cli/main.go ollama help + -disable-local-llms flag	REMOVE refs; flag removed
O8	pkg/api/handler.go "name":"ollama" models-list entry	REMOVE
O9	pkg/api/handler.go validProviders array	REMOVE ollama (and llamacpp)
O10	pkg/api/batch_handlers.go valid-provider switch	REMOVE ollama/ llamacpp arms
O11	pkg/coordination/multi_llm.go:280–283 Ollama discovery (OLLAMA_ENABLED/ OLLAMA_MODEL) + disableLocalLLMs gate (:62,94,111,139,152)	REMOVE the whole Ollama block + the disableLocalLLMs field+gates
O12	pkg/challenge_runner/provider_challenges.go ollama/llamacpp in challenge set	REMOVE both
O13	pkg/distributed/coordinator.go:231 case "ollama","llamacpp": priority arm	REMOVE/redirect per R1

Honest-boundary FACT (§11.4.6 — NOT a removal site): pkg/storage/redis.go Ollama reference is a COMMENT documenting the :-in-model-id cache-key fix; the fix code is provider-agnostic — KEEP it; reword the comment only.

1.4 SSH-remote + distributed (R1) — FACT

- **cmd/translate-ssh/main.go + cmd/ssh-translation/main.go are llama-ONLY** — grep for NewLLMTranslator/VerifiedFactory/executeAPI/bridge. in both returned EMPTY (verified). ssh-translation hardwires Type:"llamacpp", BinaryPath:"/usr/local/bin/llama.cpp". → Obsolete (R1).
- pkg/distributed/coordinator.go — local arms: getPriorityForProvider case "ollama","llamacpp" :231; local translator provisioning :297 + fallback :302; worker-provider parse comments :206,215. Redirect or remove per R1.
- pkg/distributed/pairing.go:37 — LocalLLMs []string (“ollama, llamacpp models”) worker field. REMOVE per R1.
- pkg/grpc/proto/translator.pb.go provider enum comment lists ssh/ollama/llamacpp → REGENERATE proto (never hand-edit .pb.go).

- `version_manager.go` / `ssh_pool.go` are GENERIC (no llama/ollama/gguf refs) — survive if R1=redirect; Obsolete only if R1=wholesale-remove distributed mode.

1.5 Worker configs — 8 (FACT, draft listed 5)

`config_ollama.json`, `config.distributed.json`,
`config.distributed.thinker.json`, `config.distributed.test.json`,
`config.worker.json`, `config.production.template.json`,
`config.worker.llamacpp.json`, `config.worker.ollama.json` — all under `internal/working/`, all carry a `local-provider/gguf/model_path/11434/ollama` marker. Audit each: DELETE the worker/distributed local fixtures; for shared templates (e.g. `config.production.template.json`) STRIP the local arm rather than delete.

1.6 Helper scripts — 15 (FACT, draft listed 4)

`internal/scripts/`: `batch_translate_llamacpp.sh`,
`batch_translate_markdown_llamacpp.sh`, `check_llamacpp.sh`,
`install_llamacpp.sh`, `llm_translation.sh`,
`monitor_llamacpp_translation.sh`, `setup_llama.py`,
`test_distributed_llamacpp.sh`, `test_llama.py`, `translate_llm_only.py`,
`translate_markdown_llamacpp_production.sh`,
`translate_markdown_llamacpp.sh`, `translate_markdown_multillm.sh`,
`translate_with_llama.py`, `translator-ssh-llama`. All feature-removed (R1 — local-only tooling).

1.7 Test cluster (rewrite/remove under §11.4.43/§11.4.115)

llama: `llamacpp_test.go`, `llamacpp_provider_test.go`,
`llamacpp_buildargs_defect_test.go` + arms in `llm_test.go`,
`provider_clients_test.go`, `integration_test.go`, `cmd/unified-
translator/ssh_command_test.go`, `cmd/cli/main_test.go`, `cmd/markdown-
translator/main_test.go`. ollama: `ollama_test.go`,
`ollama_options_defect_test.go` + arms in the above + `cmd/translate-ssh/
main_test.go`, `pkg/coordination/multi_llm_test.go`/
`multi_llm_fault_test.go`, `pkg/distributed/*_test.go`, `pkg/translator/
universal_test.go`, `test/unit/coordination_test.go`. KEEP `pkg/storage/
*_cachekey*_test.go` (cache-key tests — reword ollama example only).

2. Drop-in confirmation + the ONE prerequisite gap

Interfaces (FACT): - `llm.LLMClient` (`pkg/translator/llm/llm.go:211`):
`Translate(ctx, text, prompt) (string, error) + GetProviderName() string`. - `translator.Translator` (`pkg/translator/translator.go:36`):
`Translate(ctx, text, context) (string, error) + TranslateWithProgress(...) + GetStats() + GetName()`. **No**
GetProviderName(). - `bridge.BestTranslator(ctx, task) (translator.Translator, []string, error)` — backed by

VerifiedFactory.CreateTranslatorWithFallback → *llm.LLMTranslator (pkg/translator/llm/verified_factory.go:108). *LLMTranslator is the SAME concrete type NewLLMTranslator returns (llm.go:217).

Verdict — the 15 NewLLMTranslator call sites: clean drop-in. They consume the high-level Translate/TranslateWithProgress surface (the Translator contract), which bridge.BestTranslator satisfies via the identical *LLMTranslator. Replace llm.NewLLMTranslator(config) with bridge.BestTranslator(ctx, task) (default provider becomes the strongest verified model, not a hardcoded string).

Prerequisite gap P-1 (the ONLY hand-wave-free blocker) — two llm.LLMClient consumers: - pkg/markdown/simple_workflow.go:22 — field LLMPProvider llm.LLMClient. - pkg/modelsbridge/bridge.go:29,35 — wraps llm.LLMClient, calls GetProviderName() (:43,103).

bridge.BestTranslator returns translator.Translator, which lacks GetProviderName() — so it is NOT assignable to an llm.LLMClient field.

Resolution (tracked work item before R-1 touches these two): add a tiny adapter exposing Translate(ctx, text, prompt) + GetProviderName() over the bridge — source the provider name from bridge.BestModel(ctx, task).ProviderID. Either (a) bridge exposes a BestClient(ctx, task) (llm.LLMClient, error) helper, or (b) a thin local adapter in each consumer wraps BestTranslator+BestModel. Recommended: **(a)** one bridge.BestClient helper (single, tested, reused) — keeps the two consumers a clean drop-in and avoids duplicate adapters. This is a small additive bridge method; it is a PREREQUISITE for redirecting those two files, not a blocker for the other 13 sites.

3. Contention map + parallelization

Removal/redirect steps and the files each touches, ordered so they don't collide (§11.4.58/§11.4.119 — one owner per file). Disjoint-file steps parallelize; same-file steps serialize.

Step	Files (exclusive owner)	Parallel-safe with
P-1 (bridge BestClient adapter)	pkg/bridge/bridge.go (+ test)	everything (additive, no removal)
R-1a redirect cmd binaries	cmd/cli/main.go, cmd/markdown-translator/main.go, cmd/preparation-translator/main.go, cmd/unified-translator/main.go, cmd/ebook-translator/ebook_translator.go	R-1b, R-1c (disjoint files)
R-1b redirect api/grpc	pkg/api/handler.go, pkg/api/batch_handlers.go, pkg/grpc/core_translator.go	R-1a, R-1c
R-1c redirect pkg coordinators	pkg/coordination/multi_llm.go, pkg/preparation/coordinator.go,	R-1a, R-1b

Step	Files (exclusive owner)	Parallel-safe with
	pkg/verification/ polisher.go	
R-1d redirect LLMClient consumers (needs P-1)	pkg/markdown/ simple_workflow.go, pkg/ modelsbridge/bridge.go	after P-1; disjoint from R-1a/b/c
R-2 remove Ollama	pkg/translator/llm/ {ollama.go, llm.go}, lists in handler/batch (SHARED with R-1b → SERIALIZE after R-1b), provider_challenges.go, multi_llm.go (SHARED with R-1c → SERIALIZE after R-1c)	NOT parallel with R-1b/R-1c on shared files
R-3 remove llama.cpp	llamacpp.go, llamacpp_provider.go, llm.go (SHARED with R-2 → SERIALIZE), proto regen, pkg/ hardware (§11.4.124 investigate- first, separate commit)	serialize with R-2 on llm.go
R-4 SSH/distributed (R1)	cmd/translate-ssh, cmd/ssh- translation, pkg/ distributed/ {coordinator.go, pairing.go}, worker configs, scripts	distributed files disjoint from R-2/R-3 except coordinator.go (already touched in R-1c/R-2 — serialize)
R-5 gate + retest	new gate script, docs, .golangci.yml, session- resumption file	last

Serialization hot files (MUST be single-owner across steps): pkg/translator/llm/llm.go (R-2 + R-3), pkg/api/handler.go (R-1b + R-2 lists), pkg/coordination/multi_llm.go (R-1c + R-2/O11), pkg/distributed/coordinator.go (R-1c + R-2/O13 + R-4). Assign each to ONE agent across its steps, or sequence them.

What each removal BREAKS (must be edited in lockstep): the factory switch case ProviderOllama/ProviderLlamaCpp (llm.go), validProviders arrays (handler.go, batch_handlers.go), proto enum (translator.pb.go → regen), LlamaCppProviderConfig type (referenced by cmd/translate-ssh, cmd/translator :36 — both removed/redirected in R-4), the §11.4.153 feature-ledger rows for llama.cpp/Ollama/SSH (→ §11.4.90 Obsolete, §5), and the test cluster (§1.7).

4. CM-NO-LOCAL-RUNTIME gate + paired §1.1 mutation

Pre-build gate CM-NO-LOCAL-RUNTIME (generalises bridge-doc CM-NO-LLAMACPP):
- **Scans** cmd/ + pkg/ + internal/*.go and internal/working/*.json, **EXCLUDING** submodule trees (llms_verifier/, helix_qa/, vision_engine/), the removal-history test files, and the pkg/translator/llm/mock.go unit seam (mocks permitted in unit tests only, §11.4.27). - **FAILs** on any occurrence of the forbidden set: llama.cpp · llamacpp · llama-cli · ProviderLlamaCpp · ProviderOllama · OllamaClient · LlamaCppProviderConfig · :11434 / 11434 · OLLAMA_ENABLED · OLLAMA_MODEL · -disable-local-llms · (R1=A) executeSSTranslation / translate-ssh / ssh-translation / LocalLLMs. - **JSON config sub-assert:** no tracked internal/working/*.json carries "default_provider":"llamacpp" | "ollama" | gguf | model_path | n_gpu_layers | 11434. - **Paired §1.1 mutation:** re-introduce one forbidden token (e.g. re-add the ProviderOllama const in llm.go) → assert gate FAILs; restore → assert PASS. (gate-code = separate work item.)

No-key SKIP guard (R2, §11.4.3): a test asserting that with NO *_API_KEY set, every translate entry point returns an honest hard error / SKIP-with-reason — NEVER a local fallback. The bridge ALREADY enforces this (bootstrapInProgress returns “no provider API keys set ... (local llama.cpp fallback is not permitted)”, bridge.go:158–163) — the guard pins that behaviour. Paired mutation: re-add a local fallback in any redirected entry point → guard FAILs.

Per-component §11.4.115 RED-first redirect tests: per redirected component, RED_MODE=1 reproduces “component still provisions a local client” on the pre-fix artifact; flipped RED_MODE=0 is the standing guard asserting it provisions via bridge.BestTranslator. Register into the §11.4.135 regression suite.

5. R2 — require-keys-everywhere map (honest hard-fail, §11.4.69)

Every entry point that must hard-fail (no silent local fallback) when no API key + no verified model is available:

Entry point	Current no-key behaviour	Post-R2 behaviour
pkg/bridge Open/ bootstrapInProgress	already hard-errors (bridge.go:158–163)	unchanged — this IS the canonical honest path
cmd/unified-translator/ main.go:702 case "mock","ollama"	falls back to mock/ ollama no-key arm	drop ollama; redirect default to bridge → bridge hard-errors if no key
cmd/cli/main.go:350	provider+-disable- local-llms	bridge path → hard- error if no key; remove local flags

Entry point	Current no-key behaviour	Post-R2 behaviour
cmd/markdown-translator/ main.go:314	deepseek default needs key	bridge path → hard- error if no key
cmd/preparation- translator/main.go:175	resolves *_API_KEY, empty = honest	bridge path → hard- error if no key (already honest)
cmd/ebook-translator/ ebook_translator.go	PROVIDER: llamacpp detection	bridge path; remove llamacpp detection
pkg/api/handler.go:778 createTranslator	provider list	bridge path → 4xx/ honest error if no key
pkg/api/ batch_handlers.go:123,235	provider switch	bridge path → honest error
pkg/grpc/ core_translator.go:320,361	provider routing	bridge path → gRPC error status if no key
pkg/coordination/ multi_llm.go:171	per-provider + Ollama discovery	bridge path; remove Ollama discovery + disableLocalLLMs
pkg/preparation/ coordinator.go:98, pkg/ verification/ polisher.go:124	per-provider	bridge path → honest error
pkg/distributed/ coordinator.go:297,302	local translator provisioning	R1: remove (A) or redirect worker body to verified API (B)

CI with no keys ⇒ integration/e2e/Challenge suites SKIP-with-reason (§11.4.3); unit tests use mock.go (§11.4.27). No re-introduced silent local fallback (would re-violate the mandate).

6. §11.4.122 / §11.4.90 — Obsolete entries to create (operator-authorized via D1/D2/R1)

Each removal is an operator-authorized capability drop (D1 remove local runtimes; D2 forbid both; R1 Obsolete SSH/distributed). Create these §11.4.90 Obsolete tracker entries (status Obsolete (→ Fixed.md)), each with Obsolete-Details (Since / Reason / Superseding-item=pkg/bridge / Triple-check evidence = this doc's §1):

Item	Reason	Gate
cmd/translate-ssh (llama-only, grep-empty for API path)	superseded-by- design-change	R1
		R1

Item	Reason	Gate
cmd/ssh-translation (Type:"llamacpp" hardwired)	superseded-by- design-change	
pkg/translator/llm/ llamacpp.go, llamacpp_provider.go	superseded-by- design-change	always (after §11.4.124 git-history)
pkg/translator/llm/ ollama.go	superseded-by- design-change	always (after §11.4.124 git-history)
8 worker/distributed configs (§1.5)	feature-removed	worker/distributed = always; templates = strip- local-arm
15 internal/scripts/ *llama*/*llm*/ multillm/translator- ssh-llama (§1.6)	feature-removed	R1
pkg/hardware (imported only by llamacpp.go)	dead-code candidate	§11.4.124 investigate- FIRST, separate removal- only commit citing git- history
feature-ledger rows (§11.4.153) for llama.cpp / Ollama / SSH-worker	feature-removed	with their removal commits

NOT Obsolete (live non-local consumers — KEEP): pkg/models (imported by cmd/server, handler.go, user_auth.go), pkg/distributed/version_manager.go + ssh_pool.go (generic; Obsolete only if R1=wholesale distributed removal), pkg/storage/redis.go cache-key fix (provider-agnostic), cmd/model-bridge (the bridge CLI).

7. Ordered execution checklist (the implementation stream runs this)

Pre-req MET: pkg/bridge exists with BestTranslator/Invoke/BestModel/ListVerified (verified).

- **P-0 — Operator gate.** R1 (distributed/SSH remove vs redirect-worker-to-API) + R2 confirm (no-key → SKIP, no silent fallback) — both already granted via D1/D2/R1; confirm R1=A (remove) vs R1=B (redirect) before R-4. (§11.4.66/§11.4.101/§11.4.122.)
- **P-1 — Add bridge.** **BestClient(ctx, task) (llm.LLMClient, error) adapter** (additive, no removal). Tested. Unblocks R-1d. (Closes the only drop-in gap.)
- **R-1 — Redirect (additive; locals still present; builds stay green).** R-1a cmd binaries · R-1b api/grpc · R-1c pkg coordinators (parallel — disjoint files) → then R-1d the two LLMClient consumers (after P-1). Each component: §11.4.115 RED-first redirect test.

- **R-2 — Remove Ollama** (§11.4.124 git-history first; separate commit). O1–O13; reword `redis.go` comment; rewrite/remove ollama tests. Serialize on `llm.go/handler.go/multi_llm.go/coordinator.go`.
- **R-3 — Remove llama.cpp** (§11.4.124 + §11.4.122). 16 files; regenerate proto; delete `LlamaCppProviderConfig`; investigate `pkg/hardware` orphaning (separate removal-only commit with git-history evidence).
- **R-4 — SSH/distributed per R1.** (A) Obsolete the 2 SSH binaries + distributed local arms + 8 configs + 15 scripts + `-provider ssh`; OR (B) keep distributed, redirect worker body to verified API (build the API-worker body — does not exist today). Track each as §11.4.90 Obsolete.
- **R-5 — Gate + full retest.** Add CM-N0-LOCAL-RUNTIME + paired mutation + no-key SKIP guard + worker-config JSON gate. Update Documentation/AGENTS.md, README, `.golangci.yml`, the §11.4.131 session-resumption file. Full-suite retest (§11.4.40). docs/qa/<run-id>/ transcript for the bridge-backed translate path (§11.4.83).

8. §11.4.145 8-angle impact GO/NO-GO (re-confirmed)

Angle	Finding (FACT)	Verdict
1 Correctness	Bridge returns the same *LLMTranslator; deleted factory branches → unknown-provider errors via existing default arm.	GO
2 Regression-defaults	<code>config.json</code> <code>default_provider:"openai"</code> unaffected; 8 worker/distributed configs default local → delete/strip (§1.5).	GO (with config delete)
3 Latent/contract	factory switch, <code>validProviders</code> , proto enum, <code>LlamaCppProviderConfig</code> consumers all need lockstep edit (§3).	GO (lockstep)
4 Security	removes exec of llama-cli + SSH remote-binary install/upload — net RCE/supply-chain reduction.	GO (improves)
5 Performance	all inference → network API; bridge fallback chain mitigates.	GO
6 Host/data safety	removes host subprocess + remote install (§11.4.133-adjacent).	GO (improves)
7 Cross-feature	<code>version_manager</code> generic (survives R1=B); redis cache-key fix kept; <code>OLLAMA_ENABLED</code> contract removed.	GO

Angle	Finding (FACT)	Verdict
8 Business-logic	tests assume locals (§1.7); no-key CI → SKIP (R2, bridge already hard-errors).	GO (R2 honest)

Overall: GO. R1 choice (A vs B) is the only operator gate before R-4; both pre-authorized in principle.

Sources verified 2026-06-15 (§11.4.150 — carried from the companion draft)

- <https://www.sitepoint.com/hybrid-cloudlocal-llm-the-complete-architecture-guide-2026/>
- <https://www.getmaxim.ai/articles/top-5-llm-gateways-in-2025-the-definitive-guide-for-production-ai-applications/>
- <https://dev.to/akarshc/how-to-test-llm-integrations-in-ci-without-burning-tokens-1ibh>
- <https://developers.redhat.com/articles/2026/05/27/how-we-built-integration-testing-fast-moving-ai-backend>